# Upgrading FinRobot: What Happens When You Swap GPT for Claude, Gemini, and Perplexity in an Equity Research Pipeline

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*GR5398 · Spring 2026 · Assignment 2 — Model Upgrade & Equity Research Enhancement*

## The Question

FinRobot ships with a default LLM backend. What happens if you replace it



"Claude-inspired" enhancement to the research workflow look like in practice?

This post documents 20 equity research reports generated across four experiment configurations on five companies (NVDA, AMD, INTC, AAPL, GOOGL), and what we learned from comparing them.

## What We Built

### The Pipeline

We restructured FinRobot into a **four-stage multi-agent pipeline**:

```
Stage 0 (Track B): Investment Thesis Agent  ← new, Claude-inspired
Stage 1:           Expert Investor
Stage 2:           Data Analyst
Stage 3:           Critic Reviewer
```

Each agent runs sequentially, with each stage's output passed as context to the next. A key engineering fix: **agents are re-initialized per experiment** using factory functions rather than mutating `llm_config` directly — AutoGen caches config internally, so the naive approach of `agent.llm_config = new_config` does not reliably switch the underlying model.

### Track B Enhancement: Investment Thesis Agent

The most significant structural change was adding a **Stage 0 Investment Thesis Agent**, inspired by Anthropic's equity-research workflow design. Before the Expert begins free-form analysis, this agent generates a structured brief containing:

- **Bull Case** (3 specific points with metrics)
- **Bear Case** (3 specific points with metrics)
- **Key Catalysts** (time-bound, next 6–12 months)
- **Earnings Surprise Analysis** (eps_actual vs eps_estimate)
- **One-Line Thesis Summary**

This seeding step matters because it forces explicit bull/bear duality before analysis begins, provides the Expert with a structured frame rather than a blank slate, and creates a reference document the Critic can check the final report against for logical consistency.

## Enriched Financial Metrics

We expanded `compute_metrics()` from 3 indicators to 23, adding:

- Valuation: Forward PE, PEG, P/B, P/S, EV/EBITDA
- Profitability: Gross/Operating/Net margins, ROE, ROA
- Cash flow: FCF, Debt/Equity, Current Ratio
- Earnings Surprise: eps_estimate vs eps_actual, surprise %
- Analyst consensus: target price, rating mean, number of analysts

## Four Experiment Configurations

Config Expert Analyst Critic Claude_Only Claude sonnet-4.6 Claude Claude Hybrid_Claude_Gemini Claude Gemini 2.5 Pro Claude Gemini_Only Gemini Gemini Gemini Hybrid_Perplexity Perplexity sonar-pro Claude Claude

All models routed through OpenRouter for unified API access.

## Results: What the Data Shows

We scored every final report on four quantitative dimensions and found clear differences across configurations.

### Analytical Depth (Word Count)

Configuration Avg Words vs. Gemini_Only Claude_Only 4,339 +215% Hybrid_Perplexity 3,575 +160% Hybrid_Claude_Gemini 3,075 +123% Gemini_Only 1,377 baseline

Gemini-Only produced reports roughly one-third the length of Claude-Only. This is not necessarily a problem — concision has value — but reading the actual reports confirms the gap reflects missing analysis rather than efficient compression.

### Numerical Specificity

This metric counts concrete figures ($, %, B, M, x multiples) cited in each report — a proxy for how grounded the analysis is in actual data.

Configuration Avg Numeric References Claude_Only 191.2 Hybrid_Perplexity 153.6 Hybrid_Claude_Gemini 92.0 Gemini_Only 21.8

The gap between Claude_Only (191) and Gemini_Only (21.8) is nearly **9x**. Gemini's reports were stylistically fluent but relied heavily on qualitative language where Claude would cite a specific figure.

### Risk Coverage

Configuration Avg Risk Sentences Hybrid_Perplexity 35.0 Claude_Only 41.8 Hybrid_Claude_Gemini 40.5 Gemini_Only 15.6



Gemini_Only's risk coverage was less than half that of other configurations, suggesting a more optimistic default analytical tone.

## Hallucination Rate

We flagged reports where the Expert stage explicitly acknowledged that the provided financial metrics were inconsistent with reality — but continued the analysis anyway.

Configuration Hallucination Flags Gemini_Only **5/5** Claude_Only 4/5 Hybrid_Claude_Gemini 3/4 Hybrid_Perplexity **2/5**

This was the most striking finding. Gemini flagged data problems in every single report, yet never corrected them. The NVDA Expert explicitly wrote:

> *"The P/E ratios (40.8 TTM, 17.8 FWD) are inconsistent with the market. The actual TTM P/E is closer to ~75x, and the forward P/E is in the ~40x range."*

…and then proceeded to build the entire valuation analysis on the incorrect figures.

Perplexity had the lowest hallucination rate — likely because its search capability allowed it to verify metrics against live data rather than pattern-match against training data.

## The Perplexity Advantage: Real-Time Grounding

Hybrid_Perplexity was the only configuration with meaningful access to information from 2025–2026. Four out of five Thesis stage outputs contained explicit search citations, with dates confirming real-time retrieval.

The INTC report is the clearest example. The Perplexity Expert opened with:

> *"Q1 2026 delivered a transformative beat with $0.29 adjusted EPS versus $0.01 consensus — a 2,100%+ surprise — driven by explosive server CPU demand for AI inference. Management indicated unfilled demand 'starts with a B,' while DCAI margins expanded sharply to 30.5%."*

This level of recency is not possible from a model operating purely on training data. For GOOGL, the thesis referenced Alphabet's specific $175–185B capex commitment for 2026 — a figure that was disclosed after most models' training cutoffs.

The trade-off: Perplexity's Expert stage was notably shorter (avg ~4,300 chars) than Claude's (~18,000 chars), suggesting it optimized for factual retrieval over analytical depth. The architecture compensated for this by having Claude handle the Analyst and Critic stages, where depth and rigor matter most.

## The Claude Critic: When Self-Review Actually Works

The Critic Agent's job is to review the Structured Report for logical inconsistencies, missing risks, and unsupported claims — and return a corrected final version.

In the Claude_Only AAPL report, the Critic caught a real structural flaw:

> *"The primary growth driver (Services margin accretion) was presented without cross-referencing the regulatory compression risk that directly threatens it. A forward reference to Risk 2 has been embedded in the Services driver narrative."*

In the Hybrid_Perplexity AAPL report, the Critic made 18 distinct documented corrections, including:

- Flagging the +15.7% revenue growth figure as inconsistent with Apple's reported FY2024 ~2% YoY
- Raising the WACC from 7.5% to 8.0% to reflect the current rate environment
- Adding the Google search revenue arrangement (~$18–20B/year) as a missing regulatory risk
- Correcting share count reduction magnitude from "~20%" to "~40%"
- Revising bear-case probability upward from 25% to 30%, reducing expected value from $297 to $288–292

This is qualitatively different from stylistic editing — these are substantive analytical corrections that change the investment conclusion.

## What Didn't Work

**Gemini as sole pipeline operator**: Beyond the hallucination and brevity issues, Gemini_Only had a fundamental data consistency problem. In the GOOGL report, market cap, revenue, and net income figures were mutually contradictory across the three stages — each agent derived its own numbers rather than inheriting from shared context. No investment decision should be made from a report where the same company has three different revenue figures across three pages.

**Network reliability with Gemini 2.5 Pro**: The Hybrid_Claude_Gemini NVDA report failed entirely due to a 502 error from the Gemini API via OpenRouter. Gemini 2.5 Pro has higher latency and lower availability than Claude or Perplexity sonar-pro at current load levels. For production pipelines, retry logic or a fallback model is essential.

**All configurations struggled with stale yfinance data**: Even Claude, despite flagging inconsistencies, could not correct them without search access. Four out of five Claude_Only Expert stages noted metric anomalies. The enriched `compute_metrics()` function helps by providing more data points, but cannot solve the fundamental problem of training-data staleness.

## Configuration Recommendations

**For highest factual accuracy:** `Hybrid_Perplexity` Use Perplexity as the Expert to ground the analysis in current data, Claude as Analyst and Critic for depth and rigor. Best for any company where recent earnings, management guidance, or macro events are material to the thesis.

**For most consistent depth without search dependency:** `Claude_Only` Highest word count (4,339 avg), highest numeric specificity (191 refs), zero empty stages, zero network errors. Best choice when financial metrics are pre-validated externally or when API reliability is a priority.

**Avoid for standalone use:** `Gemini_Only` Shortest reports (1,377 avg words), lowest numeric specificity (21.8 refs), highest hallucination rate (5/5), and inability to produce final ratings consistently. May be useful as a fast-draft tool in a hybrid setup where another model handles critique.

## Key Takeaway

The most important finding is not about which model is "best" in isolation — it is that **model role assignment matters as much as model selection.** Perplexity performing poorly as a solo analyst is a different statement from Perplexity performing well as a search-grounded Expert with Claude handling structured output and critique. The multi-agent architecture creates a division of labor where each model's specific capability — search grounding, structured output, rigorous self-correction — can be applied at the stage where it adds the most value.

The Claude-inspired Investment Thesis Agent (Stage 0) made a measurable difference to report quality by forcing explicit bull/bear framing before analysis began, giving the Critic a reference document to check final conclusions against, and surfacing earnings surprise data that would otherwise be buried in the financial metrics dictionary.

The full pipeline, all 20 reports, and comparison code are available in the accompanying notebook.

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